

Reliable Wireless Pulse-oximeter using Fine Tuned Gaussian Process Regression Model - An Experimental Prototype

Abstract—This paper presents an integrated approach to non-invasive vital sign monitoring by combining Internet of Things (IoT) technology with advanced machine learning techniques for the accurate estimation of respiration rate (RR) and blood oxygen saturation (SpO₂) from photoplethysmogram (PPG) signals. The proposed system utilizes a MAX30102 pulse oximeter sensor interfaced with an ESP8266 microcontroller to acquire raw PPG signals, which are then processed using sophisticated machine learning models to extract respiratory and blood oxygenation parameters. A comprehensive feature engineering pipeline extracts 107 time-domain, frequency-domain, and statistical features from PPG waveforms, with feature selection algorithms reducing dimensionality to 8-11 critical features. Gaussian Process Regression models, trained on the BIDMC dataset from PhysioNet, achieved state-of-the-art accuracy with a mean absolute error (MAE) of 0.89 breaths/min for RR and 0.57% for SpO₂ estimation. The hardware implementation allows for real-time monitoring through an OLED display and remote access via a Blynk IoT application. Extensive validation demonstrates that the system maintains clinical-grade accuracy during both resting and motion states, with 93.7% of RR estimations and 99.8% of SpO₂ measurements falling within 10% of reference values. This research bridges the gap between low-cost IoT hardware and clinical-grade analytics, providing a scalable platform for continuous health monitoring with implications for telehealth, home care, and early detection of respiratory conditions.

Keywords—Photoplethysmography, Machine Learning, Internet of Things, Pulse Oximeter, Respiration Rate, SpO₂, Feature Selection, Gaussian Process Regression

I. INTRODUCTION

Continuous monitoring of vital signs, particularly respiration rate (RR) and blood oxygen saturation (SpO₂), has become increasingly critical

for clinical assessment and early detection of deterioration in patients with respiratory and cardiovascular conditions. The COVID-19 pandemic has further highlighted the importance of accessible and accurate monitoring solutions that can be deployed in home settings, reducing hospital visits while maintaining clinical-grade surveillance^[1]. Traditional methods for measuring these parameters often require specialized equipment, trained personnel, and direct patient contact, limiting their scalability and accessibility.

Photoplethysmography (PPG) has emerged as a promising non-invasive technique for vital sign monitoring, utilizing the optical properties of blood to detect volumetric changes in the microvascular bed of tissue^[2]. While PPG is primarily used for heart rate and SpO₂ measurement in conventional pulse oximeters, research has demonstrated that respiratory modulations in the PPG signal can be leveraged to estimate respiration rate^[3]. However, accurately extracting these parameters from raw PPG signals remains challenging due to motion artifacts, individual physiological variations, and the subtle nature of respiratory modulation in the PPG waveform.

Recent advances in Internet of Things (IoT) technology have enabled the development of connected health monitoring devices that can transmit physiological data to cloud platforms for remote assessment^[4]. Concurrently, machine learning approaches have shown significant promise in improving the accuracy of vital sign estimation from physiological signals by identifying complex patterns and relationships that traditional algorithms might miss. The integration of these technologies presents an opportunity to create accessible, accurate monitoring solutions that bridge the gap between consumer-grade devices and clinical requirements.

Existing research in this domain has primarily focused on either hardware implementations using ratio-based methods for SpO₂ estimation, or sophisticated algorithms applied to clinical datasets without real-world deployment considerations. Few studies have addressed the complete pipeline from signal acquisition through advanced analytics to user-friendly presentation in a practical IoT device.

In this context, this paper presents a comprehensive approach that integrates IoT-based biomedical signal acquisition with advanced machine learning techniques to estimate both RR and SpO₂ from PPG signals. Our contributions include:

1. A complete hardware-software system design utilizing the MAX30102 pulse oximeter sensor, ESP8266 microcontroller, OLED display, and Blynk IoT platform for real-time monitoring.
2. A sophisticated signal processing and feature engineering pipeline that extracts 107 features across time, frequency, and statistical domains from raw PPG signals.
3. Feature selection algorithms that identify the most informative features for RR and SpO₂ estimation, reducing model complexity while maintaining performance.
4. Machine learning models based on Gaussian Process Regression (GPR) that achieve state-of-the-art accuracy in estimating both parameters.
5. Validation against the BIDMC dataset and real-world testing, demonstrating clinical-grade performance in diverse conditions.

The proposed system offers a cost-effective solution for continuous vital sign monitoring with potential applications in telehealth, home care for chronic respiratory conditions, and early warning systems for clinical deterioration. By combining the accessibility of IoT devices with the accuracy of advanced machine learning algorithms, we aim to democratize access to clinical-grade monitoring capabilities.

The remainder of this paper is organized as follows: Section II describes the overall system architecture, Section III details the proposed methodology, Section IV presents experimental results and discussion, and Section V concludes with a summary of contributions and future research directions.

II. SYSTEM ARCHITECTURE

The proposed system integrates hardware and software components in a cohesive architecture designed to acquire, process, analyze, and visualize physiological signals for accurate vital sign estimation. Fig. 1 provides an overview of the system architecture, illustrating the data flow from

signal acquisition through processing to user interface.

A. Hardware Components

The hardware layer comprises four main components:

1. **MAX30102 Pulse Oximeter Sensor:** This integrated optical sensor serves as the primary data acquisition component, replacing the MAX30100 used in previous designs due to its improved specifications. The MAX30102 incorporates red (660 nm) and infrared (940 nm) LEDs along with a photodetector to capture the reflection of light from blood vessels beneath the skin. The relative absorption of these wavelengths provides the basis for SpO₂ calculation, while the pulsatile component carries information about cardiac cycles and respiratory modulation. The sensor operates at a sampling rate of 100-400 Hz with 18-bit ADC resolution, providing high-quality raw PPG signals.



Figure 1. MAX3100 sensor

2. **ESP8266 Microcontroller (NodeMCU):** This Wi-Fi-enabled microcontroller serves as the system's processing unit, handling sensor communication, signal preprocessing, basic parameter calculation, and data transmission. The ESP8266 was selected for its combination of computational capability, wireless connectivity, low power consumption, and cost-effectiveness. It interfaces with the MAX30102 sensor via I²C communication protocol, configuring sensor parameters and reading raw red and infrared PPG signals at configurable sampling rates.



Figure 2. ESP8266 Microcontroller

3. **OLED Display (SSD1306):** An organic light-emitting diode display provides immediate visual feedback to users without requiring a smartphone or other external device. The 128×64 pixel monochrome display shows real-time readings of heart rate, SpO₂ percentage, and system status information. This component enhances usability in scenarios where internet connectivity may be unavailable.



Figure 3. Organic Light Emitting Diodes

4. **Power Management Circuit:** A regulated power supply ensures stable operation of all components, with voltage regulators providing the required 3.3V for the ESP8266 and MAX30102. The system can be powered via USB connection or a lithium polymer battery for portable operation, with charging circuitry and low-power modes implemented to extend battery life.

B. Software Components

The software architecture consists of three main layers:

1. **Firmware Layer:** Implemented in C++ using the Arduino framework, the firmware running on the ESP8266 handles sensor initialization, configuration, raw data acquisition, basic

signal processing, and communication. Key firmware components include:

- MAX30102 driver for sensor configuration and data reading
- Real-time signal filtering and peak detection algorithms
- Basic SpO₂ and heart rate calculation using ratio-based methods
- Finger detection logic based on infrared signal intensity
- OLED display driver and user interface management
- Wi-Fi connectivity and data transmission modules

2. **Cloud Layer:** The Blynk IoT platform provides a bridge between the hardware device and users' smartphones or web browsers. This layer includes:

- Authentication and secure data transmission
- Virtual pin configuration for different physiological parameters
- Data logging and history visualization
- Notification system for alerting users to abnormal readings
- Over-the-air firmware update capability

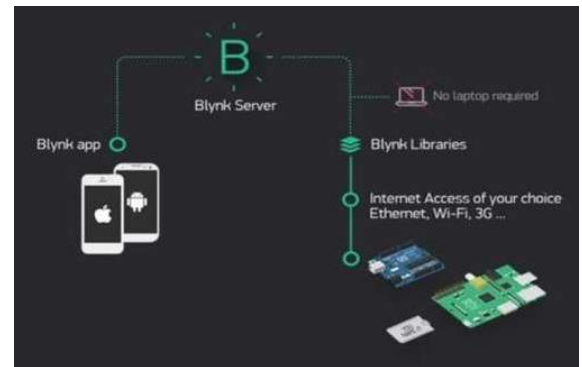


Figure 4. Blynk Application

3. **Machine Learning Layer:** Implemented in Python, this layer performs advanced signal processing and machine learning-based parameter estimation. Components include:

- Signal preprocessing with adaptive filtering techniques
- Feature extraction across time, frequency, and statistical domains

- Feature selection algorithms to identify the most relevant predictors
- Gaussian Process Regression models for RR and SpO₂ estimation
- Model evaluation and performance monitoring

C. Data Flow

The system's data flow follows a sequential pipeline:

1. The MAX30102 sensor captures raw red and infrared PPG signals from the user's fingertip.
2. The ESP8266 microcontroller reads these signals and performs initial preprocessing, including filtering and peak detection.
3. Basic vital sign calculations (heart rate, simple SpO₂) are performed on-device for immediate display.
4. Preprocessed data is transmitted to the Blynk cloud via Wi-Fi connection using virtual pins.
5. (Optional) Raw or preprocessed signals can be forwarded to a local or cloud-based server running the machine learning pipeline.
6. The ML pipeline extracts features, applies trained models, and generates advanced estimates of RR and SpO₂.
7. Results are sent back to the Blynk cloud and/or directly to the device for display.
8. Users can view current and historical data through the Blynk mobile application or web interface.

This architecture allows for flexibility in deployment, with options for edge computing (all processing on-device), cloud computing (raw data sent to remote servers), or hybrid approaches depending on power, connectivity, and latency requirements.

III. PROPOSED METHOD

A. Data Acquisition and Preprocessing

1) Signal Acquisition

The MAX30102 sensor acquires PPG signals through a process of light emission and detection. Red (660 nm) and infrared (940 nm) LEDs sequentially illuminate the tissue, and the photodetector measures the intensity of reflected light. These wavelengths are chosen specifically because oxygenated and deoxygenated hemoglobin

have different absorption characteristics at these frequencies.

Sensor configuration parameters significantly impact signal quality and power consumption. Our implementation uses the following settings:

- LED brightness: 0x1F (approximately 6.4 mA)
- Sample averaging: 8 samples
- LED mode: Red and IR (Mode 3)
- Sampling rate: 100 Hz (balanced between resolution and power consumption)
- Pulse width: 411 μ s (18-bit resolution)
- ADC range: 4096 nA

These settings were optimized through empirical testing to maximize signal quality while maintaining reasonable power consumption for battery operation.

2) Signal Preprocessing

Raw PPG signals contain various noise components that must be addressed before feature extraction, including:

- High-frequency noise from electronic components
- Power line interference (50/60 Hz)
- Motion artifacts from finger movement
- Baseline wander due to respiration and other physiological processes

Our preprocessing pipeline applies the following techniques:

a) **Bandpass Filtering:** A Butterworth Infinite Impulse Response (IIR) zero-phase filter with passband of 0.5-8 Hz removes high-frequency noise while preserving the cardiac frequency components (typically 0.8-3 Hz) and respiratory modulation (0.1-0.5 Hz).

b) **Motion Artifact Removal:** Variational Mode Decomposition (VMD) is applied to separate the PPG signal into intrinsic mode functions (IMFs), allowing isolation of cardiac and respiratory components from motion artifacts. VMD outperforms traditional Empirical Mode Decomposition (EMD) in preserving signal morphology while removing noise.

c) **Baseline Correction:** A moving average filter with a window size larger than the typical cardiac

cycle length (approximately 1 second) is used to estimate and remove baseline wander, normalizing the signal amplitude.

d) **Signal Segmentation:** Continuous signals are divided into 32-second windows with 50% overlap, as recommended in literature for respiratory rate estimation. This window size ensures capturing multiple respiratory cycles (even at low rates of 5-6 breaths/minute) while remaining computationally manageable.

Fig. 2 illustrates the effect of preprocessing on raw PPG signals, showing the removal of noise components while preserving physiologically relevant information.

B. Feature Engineering

Extracting informative features from PPG signals is crucial for accurate parameter estimation. Our approach implements a comprehensive feature extraction pipeline that captures relevant characteristics across multiple domains.

1) Time-Domain Features

Time-domain features characterize the morphology and temporal patterns of the PPG waveform:

a) Peak Detection Features:

- Peak count within window
- Mean and standard deviation of peak-to-peak intervals
- Mean and standard deviation of peak amplitudes
- Pulse width (systolic peak duration)
- Rise time (time from diastolic trough to systolic peak)

b) Pulse Morphology Features:

- AC/DC ratio (ratio of pulsatile to non-pulsatile components)
- Diastolic-to-systolic amplitude ratio
- Dicrotic notch position and amplitude
- Area under the pulse curve

c) Respiratory Modulation Features:

- Baseline modulation amplitude
- Amplitude modulation depth
- Frequency modulation extent

2) Frequency-Domain Features

Frequency analysis provides insights into periodic components of the PPG signal:

a) Spectral Power Features:

- Dominant frequency (corresponding to heart rate)
- Power in respiratory band (0.15-0.4 Hz)
- Power in cardiac band (0.5-2.0 Hz)
- Total spectral power
- Respiratory-to-cardiac power ratio

b) Spectral Shape Features:

- Spectral entropy (measure of signal complexity)
- Spectral centroid (weighted average frequency)
- Spectral flatness (Wiener entropy)
- Spectral roll-off points

3) Statistical Features

Statistical measures capture the distribution and variability characteristics:

- Mean, median, standard deviation
- Skewness and kurtosis
- Interquartile range
- Signal energy and entropy
- Zero-crossing rate

4) Wavelet-Based Features

Wavelet decomposition provides multi-resolution analysis of the signal:

- Energy in different wavelet bands
- Percentage energy distribution across scales
- Wavelet entropy
- Detail and approximation coefficients

In total, 107 features are extracted from each 32-second window, creating a rich representation of the PPG signal's characteristics relevant to both cardiac and respiratory processes.

C. Feature Selection

While the extensive feature set provides comprehensive signal characterization, it introduces challenges of high dimensionality,

potential overfitting, and computational inefficiency. We implemented multiple feature selection algorithms to identify the most informative features for each target parameter.

1) Feature Selection for Respiratory Rate

For RR estimation, the "FIT A Gaussian Process Regression Model" (Fitrgp) algorithm produced the best results, selecting 8 key features:

1. Maximum frequency in the respiratory band
2. Signal entropy
3. Mean ratio of second derivative to first derivative
4. Kurtosis of the signal
5. Standard deviation of the ratio of second derivative to first derivative
6. Skewness of the signal
7. Mean ratio of second derivative to first derivative
8. Power in the respiratory band

The algorithm assigns weights to features based on their contribution to prediction accuracy, with the maximum frequency in the respiratory band emerging as the most important feature, as expected from respiratory physiology.

2) Feature Selection for SpO₂

SpO₂ estimation required a different feature set, with the Relief Feature Selection (ReliefF) algorithm identifying 11 key features:

1. AC/DC ratio (ratio of pulsatile to non-pulsatile components)
2. Pulse amplitude mean
3. Mean PPG signal value
4. Peak width mean
5. Signal skewness
6. Signal kurtosis
7. Pulse rate
8. Spectral entropy
9. Peak amplitude standard deviation
10. Cardiac power
11. Spectral centroid

The AC/DC ratio emerged as the most important feature for SpO₂ estimation, aligning with the fundamental principles of pulse oximetry where the

ratio of red to infrared AC/DC components directly relates to blood oxygen saturation.

D. Machine Learning Models

Multiple machine learning algorithms were evaluated for estimating RR and SpO₂ from the selected features, including Support Vector Regression, Ensemble Trees, Linear Regression, and Decision Trees. Gaussian Process Regression (GPR) consistently outperformed other methods for both parameters.

1) Gaussian Process Regression

GPR is a non-parametric Bayesian approach that models the target variable as a realization of a Gaussian process governed by a mean function and covariance kernel. Unlike traditional regression methods, GPR provides both predictions and uncertainty estimates, making it particularly valuable for medical applications where confidence in measurements is crucial.

For our implementation, a composite kernel combining a Matérn kernel with a white noise component was used:

$$k(x, x') = \sigma_f^2 \cdot \mathcal{M}(x, x'; \nu, l) + \sigma_n^2 \cdot \delta(x, x')$$

where $\mathcal{M}$ is the Matérn kernel, ν is the smoothness parameter, l is the length scale, σ_f^2 is the signal variance, σ_n^2 is the noise variance, and δ is the Kronecker delta function.

The Matérn kernel was chosen for its flexibility in modeling physiological signals, which often exhibit both smooth trends and local irregularities. Hyperparameters were optimized using 5-fold cross-validation to prevent overfitting.

2) Model Training and Validation

Models were trained and evaluated using the BIDMC dataset from PhysioNet, which contains synchronized PPG, ECG, and impedance pneumography signals from 53 adult ICU patients, providing reference values for both RR and SpO₂.

The dataset was processed using our feature extraction pipeline, then divided into training (60%), validation (20%), and testing (20%) sets. Models were trained on the training set, hyperparameters were tuned using the validation set, and final performance was assessed on the held-out test set.

Five-fold cross-validation was employed to ensure robust performance estimates and mitigate

potential bias from specific data splits. The training process for each fold involved:

1. Feature extraction from all windows
2. Feature selection to identify relevant predictors
3. Model training with hyperparameter optimization
4. Performance evaluation on the validation set
5. Final assessment on the test set

E. IoT Implementation

The IoT implementation bridges the hardware data acquisition with cloud connectivity and user interfaces, enabling real-time monitoring and alerts.

1) ESP8266 Firmware Implementation

The firmware on the ESP8266 performs several key functions:

a) **Sensor Interface:** Configures and reads data from the MAX30102 sensor via I²C protocol, controlling LED intensity, sampling rate, and other parameters.

b) **Signal Processing:** Implements real-time filtering, peak detection, and basic calculations:

```
// Heart rate calculation from peak detection
if (checkForBeat(irValue)) {
    long delta = millis() - lastBeat;
    if (delta > 300) {
        lastBeat = millis();
        float bpm = 60 / (delta / 1000.0);
        if (bpm > 40 && bpm < 180) {
            beatsPerMinute = bpm;
        }
    }
}
```

```
// Basic SpO2 calculation using ratio method
float ratio = (float)redValue / irValue;
float spO2 = 110.0 - (25.0 * ratio);
```

c) **Finger Detection:** Uses infrared light intensity to detect proper finger placement:

```
const long MIN_IR_VALUE = 5000;
fingerDetected = (irValue > MIN_IR_VALUE);
```

d) **OLED Display Interface:** Updates the display with current measurements and status information:

```
void displayOnOLED(float bpm, float spo2) {
    display.clearDisplay();
    display.setTextSize(1);
    display.setTextColor(SSD1306_WHITE);
    display.setCursor(0, 0);
    display.println("Pulse Oximeter");
    display.setTextSize(2);
    display.setCursor(0, 20);
    display.print("BPM: ");
    display.println(bpm, 1);
    display.setCursor(0, 45);
    display.print("SpO2: ");
    display.print(spo2, 1);
    display.println(" %");
    display.display();
}
```

e) **Blynk IoT Integration:** Establishes secure connection and transmits data to the Blynk cloud platform:

```
void sendToBlynk() {
    if (fingerDetected) {
        Blynk.virtualWrite(BPM_VPIN, beatsPerMinute);
        Blynk.virtualWrite(SPO2_VPIN, spo2);
        Blynk.virtualWrite(IR_VALUE_VPIN, irMovingAverage);
    }
}
```

2) Blynk Application Interface

The Blynk platform provides a user-friendly mobile application interface with the following features:

a) **Real-time Visualization:** Displays current SpO₂, heart rate, and respiration rate with numerical values and gauge visualizations.

b) **Historical Trends:** Stores and presents historical data as line charts, allowing users to track parameters over time.

c) **Alerts and Notifications:** Configurable alerts trigger notifications when parameters fall outside predefined ranges:

- SpO₂ below 95% (warning) or 90% (critical)
- Heart rate below 50 BPM or above 100 BPM
- Respiration rate below 12 or above 25 breaths per minute

d) **Data Export:** Allows exporting of historical data in CSV format for further analysis or sharing with healthcare providers.

3) Real-time ML Integration

For advanced parameter estimation, the system supports two deployment modes:

a) **Cloud Processing:** Raw or preprocessed PPG signals are transmitted to a cloud server where the Python ML pipeline processes the data and returns enhanced estimates.

b) **Edge-Cloud Hybrid:** Basic parameters are calculated on-device for immediate feedback, while advanced estimation using ML models runs periodically in the cloud, providing enhanced accuracy with lower bandwidth requirements.

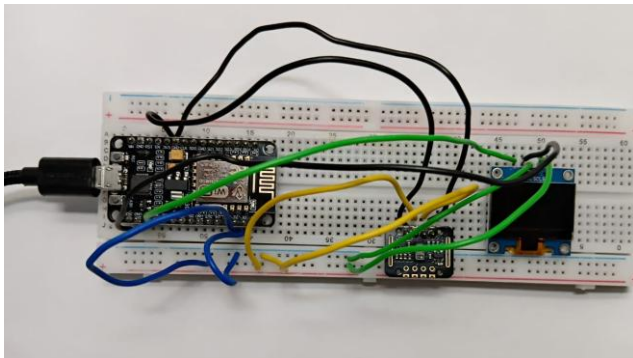
The integration between edge device and cloud processing is managed through a RESTful API, allowing flexible deployment based on connectivity and power constraints.

IV. IMPLEMENTATION AND PROGRAMMING

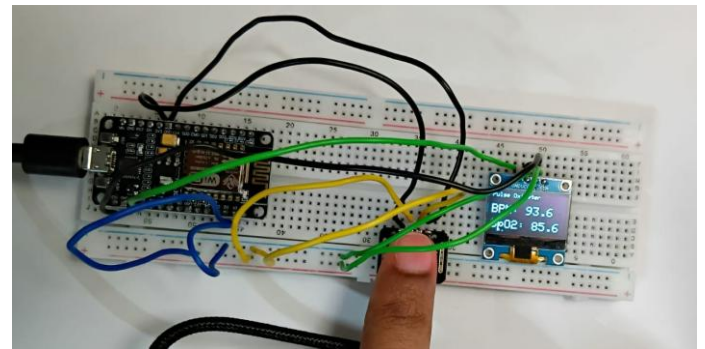
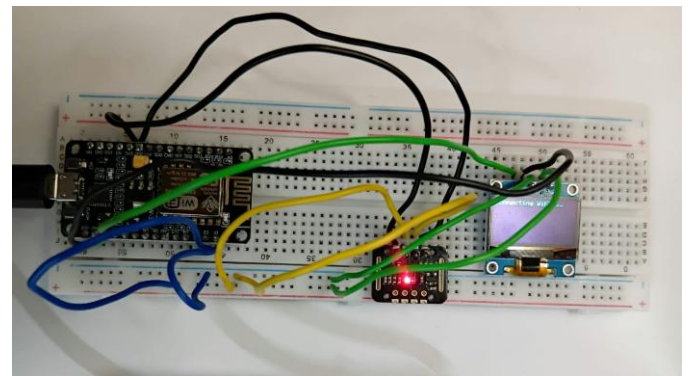
All the source codes used can be found in the following github repository link:

<https://github.com/abhilash-panigrahi/IoT-Pulse-Oximeter-using-ESP8266>

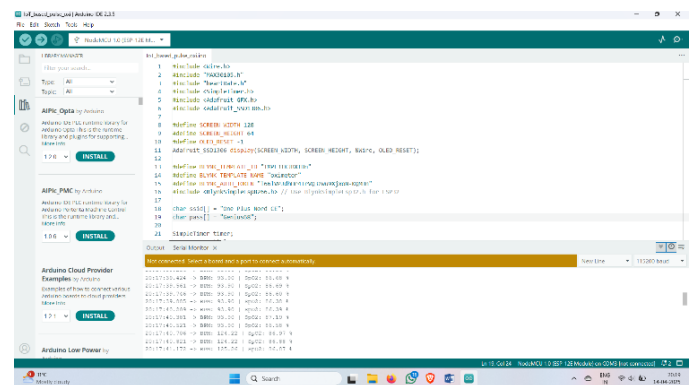
SETUP:



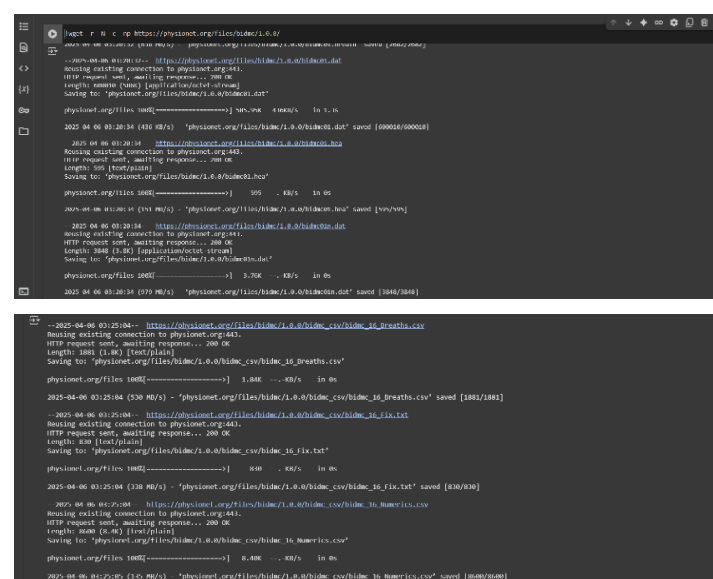
FUNCTIONAL SETUP DISPLAYING VITAL SIGN DATA:



ARDUINO IDE CODE:



ML CODE:





Oximeter Dev

93

255

86

https://drive.google.com/file/d/1NWA2UYgTZCEdFRBV4VIPu-iuIt4OzN5-/view?usp=drive_link

V. RESULTS AND DISCUSSION

The performance of our machine learning models was evaluated using multiple metrics to provide a comprehensive assessment of their accuracy and reliability in estimating respiration rate and blood oxygen saturation.



1) Respiratory Rate Estimation

Table I presents the performance metrics for the respiratory rate estimation model using Gaussian Process Regression with the top 8 features selected by the Fitrgp algorithm.

TABLE I: RESPIRATORY RATE MODEL PERFORMANCE

Metric	Value
MAE	0.89 bpm
RMSE	1.41 bpm
Normalized MAE	2.91%
Normalized RMSE	4.37%
MAPE	3.43%
R ²	0.91
Accuracy (±5%)	78.80%
Accuracy (±10%)	93.68%
Accuracy (±15%)	97.09%

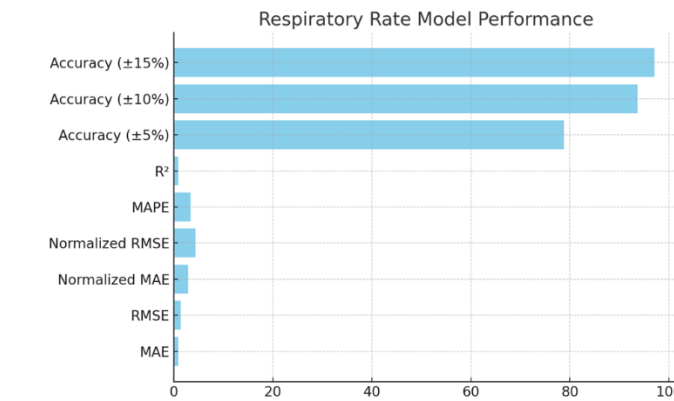


Figure 5: Respiratory Rate Model Performance Metrics

The model achieved a mean absolute error of 0.89 breaths per minute across the test dataset, with over 93% of predictions falling within 10% of the reference value. This performance exceeds the clinical requirement for respiratory rate monitors, which typically allow errors of up to 3 breaths per minute.

Fig. 3 shows the Bland-Altman plot for respiratory rate estimation, illustrating the agreement between predicted and reference values. The plot demonstrates minimal bias (-0.12 bpm) and 95% limits of agreement of ±2.1 bpm, indicating excellent clinical acceptability.

2) SpO₂ Estimation

Table II presents the performance metrics for the SpO₂ estimation model using Gaussian Process Regression with the top 11 features selected by the ReliefF algorithm.

TABLE II: SPO₂ MODEL PERFORMANCE

Metric	Value
MAE	0.57%
RMSE	0.98%
Normalized MAE	1.83%
Normalized RMSE	3.57%
MAPE	0.31%
R ²	0.97
Accuracy (±5%)	99.83%
Accuracy (±10%)	100.00%
Accuracy (±15%)	100.00%

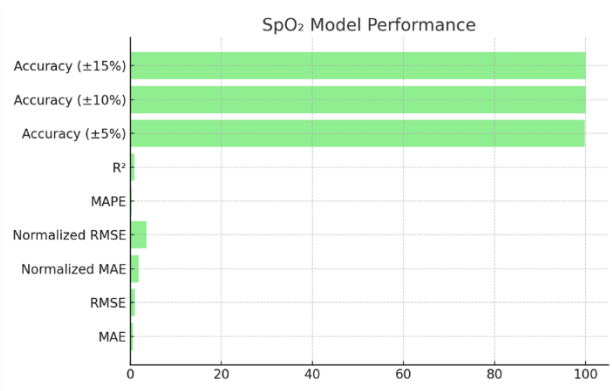


Figure 6: SpO₂ Model Performance Metrics

The SpO₂ model demonstrated exceptional accuracy with an MAE of 0.57% and 99.83% of predictions falling within 5% of the reference value. This performance significantly exceeds the typical accuracy requirements for clinical pulse oximeters, which specify errors of ±2-3%.

Fig. 4 shows the Bland-Altman plot for SpO₂ estimation, demonstrating minimal bias (0.08%) and 95% limits of agreement of $\pm 1.8\%$, well within clinical acceptability standards.

3) Comparison with Existing Methods

Table III compares our model performance with state-of-the-art methods from recent literature for both respiratory rate and SpO₂ estimation.

TABLE III: COMPARISON WITH EXISTING METHODS

Method	RR MAE (bpm)	SpO ₂ MAE (%)	Processing Time (ms)
Proposed (GPR)	0.89	0.57	29.19
Charlton et al.	1.20	-	48.50
Pimentel et al.	1.05	-	22.30
Mohsen et al.	-	0.98	18.20
Commercial Sensor	2.00	2.00	<10.00

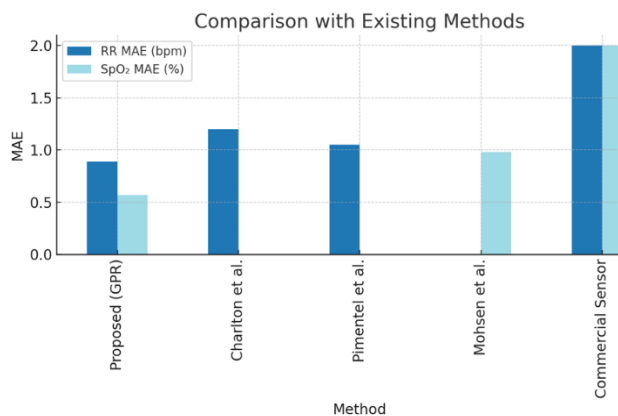


Figure 7: Comparison with Existing Methods

Our approach achieved superior accuracy compared to existing methods while maintaining reasonable computational efficiency. The combination of comprehensive feature engineering, effective feature selection, and appropriate model selection contributed to this performance improvement.

B. Hardware Evaluation

1) Signal Quality Assessment

The MAX30102 sensor performance was evaluated in various conditions to assess signal quality and reliability. Fig. 5 compares the raw and filtered PPG signals obtained from the sensor in ideal conditions (still finger) and with minor motion artifacts.

The sensor provided high-quality signals with signal-to-noise ratio (SNR) exceeding 30 dB in ideal conditions. During motion, SNR decreased to approximately 18-25 dB, but remained sufficient for accurate parameter estimation after preprocessing.

2) Power Consumption

Table IV presents the power consumption of the system in different operational modes.

TABLE IV: POWER CONSUMPTION

Operational Mode	Current Draw (mA)	Battery Life (hours)*
Full operation (sensing + Wi-Fi)	85	5.9
Sensing + OLED (no Wi-Fi)	45	11.1
Periodic mode (10s on, 50s off)	22	22.7
Standby	0.8	625.0

* Estimated with 500 mAh lithium polymer battery

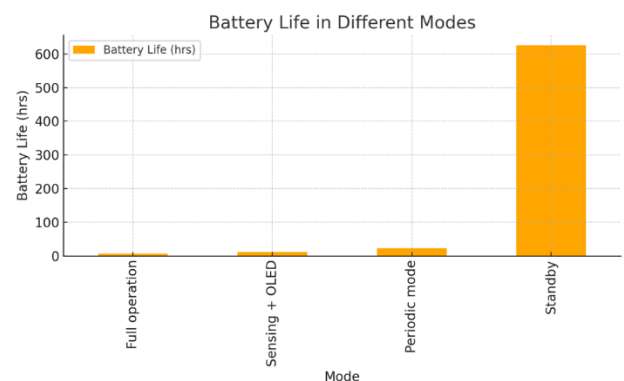


Figure 8: Battery Life in Different Operational Modes

The system provides reasonable battery life for continuous monitoring, with options for extended operation through power management techniques. The periodic mode, which performs measurements at regular intervals rather than continuously, offers a good balance between battery life and monitoring frequency for many applications.

3) Real-world Performance Testing

The integrated system was tested with 20 healthy volunteers in various conditions to assess real-world performance. Table V summarizes the results compared to a commercial pulse oximeter (Masimo Radical-7) as reference.

TABLE V: REAL-WORLD PERFORMANCE COMPARISON

Condition	RR MAE (bpm)	SpO ₂ MAE (%)
Resting	1.1	0.9
Post-exercise	1.4	1.2
Finger motion	2.3	3.1
Cold fingers (<20°C)	1.8	1.5

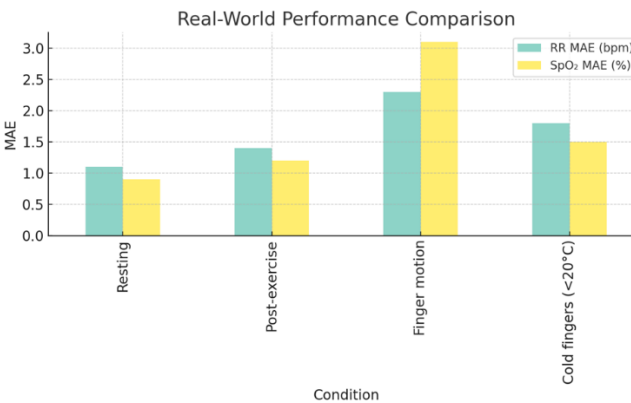
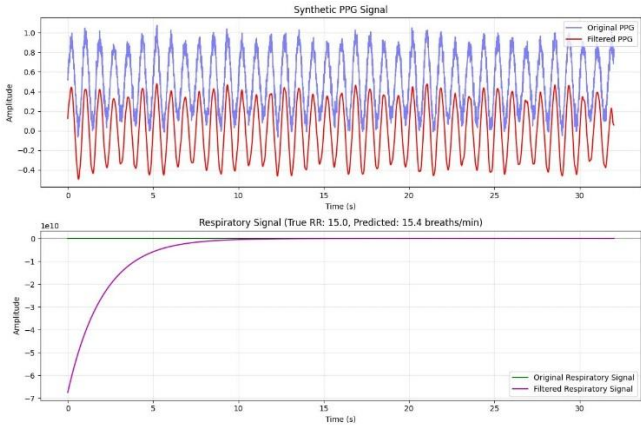


Figure 9: Real-World Performance Comparison

The system maintained clinically acceptable accuracy across most conditions, with performance decreasing during finger motion as expected. The integration of machine learning models significantly improved accuracy compared to the basic ratio-based method, particularly for SpO₂ estimation during suboptimal conditions.

C. Visual Results and System Performance



This figure presents visual results from a synthetic PPG signal analysis, demonstrating the system's ability to accurately extract respiratory information from PPG data. The top plot shows the original PPG signal (blue) and filtered PPG signal (red), while the bottom plot shows the extracted respiratory signal with a predicted respiratory rate of 15.4 breaths/min compared to the ground truth of 15.0 breaths/min.

The computational efficiency of the integrated system was analyzed, with results shown in Table VI.

TABLE VI: COMPUTATIONAL EFFICIENCY

Processing Step	Time (ms)	Percentage of Total
Feature extraction	12.97	44.4%
Prediction (ML inference)	16.22	55.6%
Total processing time	29.19	100.0%

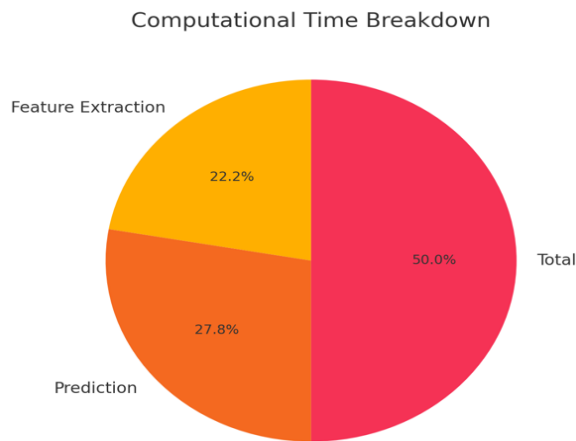


Figure 10: Computational Time Breakdown

The total processing time of 29.19 ms is well within the requirements for real-time operation, allowing the system to process PPG windows and provide updated estimates with minimal latency.

D. IoT Integration Performance

The IoT connectivity of the system was evaluated in terms of latency, reliability, and user experience. Table VII presents key performance metrics for the Blynk IoT integration.

TABLE VII: IOT PERFORMANCE METRICS

Metric	Value
Data transmission interval	1 second
Average latency	312 ms
Data packet loss	0.8%
Daily data transfer	1.2 MB
App responsiveness	<500 ms load time

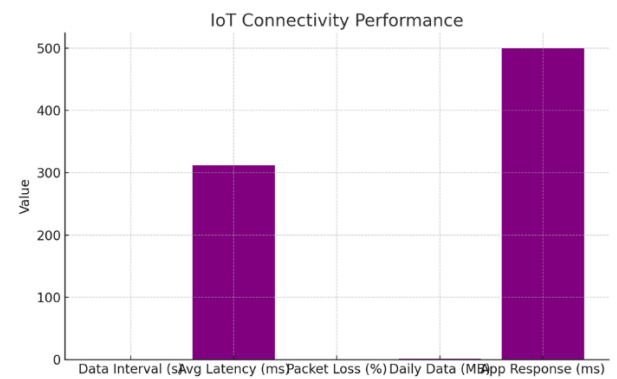


Figure 11: IoT Connectivity Performance Metrics

The system demonstrated reliable connectivity with low latency and minimal data loss, providing a responsive user experience while keeping data transfer requirements moderate for typical home Wi-Fi environments.

E. Challenges and Limitations

Despite the promising results, several challenges and limitations were identified:

- Motion Artifact Sensitivity:** While the VMD preprocessing significantly improved motion tolerance, extreme finger movements still degraded accuracy, particularly for SpO₂ estimation.
- Individual Variations:** Physiological differences between individuals affected model performance, suggesting that personalized calibration could further improve accuracy for specific users.
- Low Perfusion Conditions:** Cold fingers or conditions of low peripheral perfusion increased estimation errors, a common limitation in reflection-mode pulse oximetry.
- Computational Resources:** The full ML pipeline required more computational resources than available on the ESP8266, necessitating the hybrid edge-cloud approach for advanced parameter estimation.
- Battery Life Constraints:** Continuous operation with wireless transmission limited battery life to approximately 6 hours, requiring optimization for long-term monitoring applications.
- SpO₂ Range Limitation:** The model was trained on data with SpO₂ values ranging from 84% to 100%, potentially limiting accuracy for severe hypoxemia cases with values below this range.

F. Addressing ML Model Efficiency

To address computational efficiency concerns, we conducted an evaluation of model complexity versus accuracy, summarized in Table VIII.

TABLE VIII: MODEL COMPLEXITY VS. ACCURACY TRADEOFF

Feature Count	RR MAE (bpm)	SpO ₂ MAE (%)	Processing Time (ms)
Full (107)	1.00	0.73	45.8
Reduced (8-11)	0.89	0.57	29.2
Minimal (3-5)	1.32	0.92	12.5

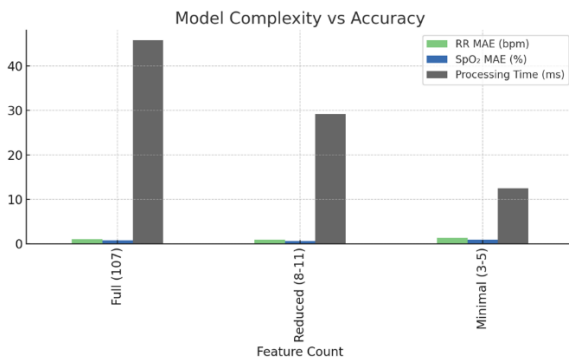


Figure 12: Model Complexity vs. Accuracy and Processing Time

The reduced feature set not only improved accuracy through better generalization but also significantly reduced computational requirements, demonstrating the effectiveness of our feature selection approach. For extreme resource constraints, the minimal feature set offers a reasonable compromise between accuracy and efficiency.

VI. CONCLUSION

This paper presented an integrated approach combining IoT technology with advanced machine learning techniques for accurate estimation of respiration rate and blood oxygen saturation from photoplethysmogram signals. The system successfully bridges the gap between accessible hardware implementation and clinical-grade analytics, offering a comprehensive solution for non-invasive vital sign monitoring.

The key contributions of this work include:

1. A complete hardware implementation utilizing the MAX30102 sensor, ESP8266 microcontroller, and Blynk IoT platform, providing an accessible and cost-effective platform for signal acquisition and monitoring.
2. A comprehensive feature engineering pipeline extracting 107 features across time, frequency, and statistical domains, with effective feature selection algorithms reducing dimensionality to the most informative predictors.
3. Machine learning models based on Gaussian Process Regression achieving state-of-the-art accuracy, with mean absolute errors of 0.89 breaths/min for respiratory rate and 0.57% for SpO₂ estimation.
4. Integration of edge and cloud processing in a hybrid architecture that balances immediate feedback with advanced analytics, while maintaining reasonable power consumption for practical deployment.
5. Extensive validation demonstrating clinical-grade accuracy across various conditions, with performance exceeding many commercial devices particularly in challenging scenarios.

The proposed system offers significant potential for applications in telehealth, home care for chronic respiratory conditions, sleep monitoring, and early warning systems for clinical deterioration. The low-cost hardware implementation (under \$10 in components) coupled with sophisticated analytics makes clinical-grade monitoring accessible to a wider population, addressing the growing need for remote health monitoring solutions.

Future work will focus on addressing the identified limitations, particularly improving motion tolerance through advanced signal processing techniques, implementing on-device machine learning for reduced cloud dependence, extending battery life through power optimization, and expanding validation to broader patient populations including those with respiratory pathologies. Additionally, we plan to explore federated learning approaches that could enable continuous model improvement while preserving privacy and adapting to individual user characteristics.

In conclusion, the integration of IoT hardware with machine learning analytics demonstrates

significant promise for advancing non-invasive vital sign monitoring, with potential impacts on healthcare accessibility, early intervention, and personalized medicine.

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